

# Command Line Interface CLI

- **Komplexe Command Line Tool-Chain «dotnet.exe»**
  - Teil des .NET Core SDK
  - Basis für high-level Tools (Visual Studio, Visual Studio Code, etc.)
- **Command-Struktur (ähnlich git)**  
`dotnet[.exe] <Verb> <argument> --<option> <param>`
  - `<verb>`            Auszuführende Aktion (build, run, etc.)
  - `<arg>`             Argument für vorangehendes Verb
  - `<option>`          Option / Switch Parameter. Mehrere Optionen möglich.
  - `<param>`          Parameter zur Option. Nicht zwingend.
- **Beispiele**  
`dotnet publish my_app.csproj`  
`dotnet build --output /build_output`

## Command Line Interface CLI

# Command Line Interface CLI

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| add          | Add a package or reference to a .NET project.                     |
| build        | Build a .NET project.                                             |
| build-server | Interact with servers started by a build.                         |
| clean        | Clean build outputs of a .NET project.                            |
| format       | Apply style preferences to a project or solution.                 |
| help         | Opens the reference page in a browser for the specified command.  |
| list         | List packages or references of a .NET project.                    |
| msbuild      | Run Microsoft Build Engine (MSBuild) commands.                    |
| new          | Create a new .NET project or file.                                |
| nuget        | Provides additional NuGet commands.                               |
| pack         | Create a NuGet package.                                           |
| publish      | Publish a .NET project for deployment.                            |
| remove       | Remove a package or reference from a .NET project.                |
| restore      | Restore dependencies specified in a .NET project.                 |
| run          | Build and run a .NET project output.                              |
| sdk          | Manage .NET SDK installation.                                     |
| sln          | Modify Visual Studio solution files.                              |
| store        | Store the specified assemblies in the runtime package store.      |
| test         | Run unit tests using the test runner specified in a .NET project. |
| tool         | Install or manage tools that extend the .NET experience.          |
| vstest       | Run Microsoft Test Engine (VSTest) commands.                      |
| workload     | Manage optional workloads.                                        |

## Ansätze

### Database First

- .NET Command Line Interface (CLI)

```
dotnet add package  
    Microsoft.EntityFrameworkCore.Tools  
dotnet add package  
    Microsoft.EntityFrameworkCore.Design  
dotnet add package  
    Microsoft.EntityFrameworkCore.SqlServer  
dotnet ef dbcontext scaffold  
    "<ConnectionString>"      <- woher  
    "Microsoft.EntityFrameworkCore.SqlServer"  
    -o Models <- wo models hin exportiert werden sollen (in Ordner Models)
```
- PowerShell «Scaffold-DbContext»
  - Tools > NuGet Package Manager > Package Manager Console

Einmalig generieren, danach müssen Änderungen manuell gemacht werden.

### Code First

- .NET Command Line Interface (CLI)

```
dotnet ef database update
```
- PowerShell «Update-Database»
  - Tools > NuGet Package Manager > Package Manager Console

## Commands «CLI»

| Command [dotnet ef ...] | Beschreibung                                           |
|-------------------------|--------------------------------------------------------|
| ... database drop       | Drops the database.                                    |
| ... database update     | Updates the database to a specified migration.         |
| ... dbcontext info      | Gets information about a DbContext type.               |
| ... dbcontext list      | Lists available DbContext types.                       |
| ... dbcontext scaffold  | Scaffolds a DbContext and entity types for a database. |
| ... migrations add      | Adds a new migration.                                  |
| ... migrations list     | Lists available migrations.                            |
| ... migrations remove   | Removes the last migration.                            |
| ... migration script    | Generates a SQL script from migrations.                |

<https://docs.microsoft.com/en-us/ef/core/miscellaneous/cli/dotnet>

# Commands «Package Manager»

| Command                    | Beschreibung                                                                                                |
|----------------------------|-------------------------------------------------------------------------------------------------------------|
| Get-Help EntityFramework   | Displays information about entity framework commands.                                                       |
| Add-Migration <mig-name>   | Creates a migration by adding a migration snapshot.                                                         |
| Remove-Migration           | Removes the last migration snapshot.                                                                        |
| Update-Database <mig-name> | Updates the database schema based on the last migration snapshot.                                           |
| Script-Migration           | Generates a SQL script using all the migration snapshots.                                                   |
| Scaffold-DbContext         | Generates a DbContext and entity type classes for a specified database. This is called reverse engineering. |
| Get-DbContext              | Gets information about a DbContext type.                                                                    |
| Drop-Database              | Drops the database.                                                                                         |

<https://docs.microsoft.com/en-us/ef/core/miscellaneous/cli/powershell>

# Fields / Scalar Value Types

| .proto | C#                                                                                                                                                                                                | Default | Beschreibung                                                                                                                                    |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| double | double                                                                                                                                                                                            | 0       |                                                                                                                                                 |
| float  | float                                                                                                                                                                                             | 0       |                                                                                                                                                 |
| int32  | int                                                                                                                                                                                               | 0       | Uses variable-length encoding. Inefficient for encoding negative numbers – if your field is likely to have negative values, use sint32 instead. |
| int64  | long                                                                                                                                                                                              | 0       | Uses variable-length encoding. Inefficient for encoding negative numbers – if your field is likely to have negative values, use sint64 instead. |
| bool   | bool                                                                                                                                                                                              | false   |                                                                                                                                                 |
| string | string                                                                                                                                                                                            | <empty> | A string must always contain UTF-8 encoded or 7-bit ASCII text.                                                                                 |
| bytes  | ByteString                                                                                                                                                                                        | <empty> | May contain any arbitrary sequence of bytes.                                                                                                    |
| [...]  | Weitere Ganzzahltypen wie [u s][int long][32 64] oder fixed[32 64]<br><a href="https://protobuf.dev/programming-guides/proto3/#scalar">https://protobuf.dev/programming-guides/proto3/#scalar</a> |         |                                                                                                                                                 |

neu mit empty und date

default(T)

bei initialisierung wird Speicher auf members  
schon genutzt auch bei leerem Konstruktor.

ohne angeben von Wert, wird default gesetzt

## Standard-Werte / default-Literal

| Typ     | Default                                | Typ    | Default                                   |
|---------|----------------------------------------|--------|-------------------------------------------|
| class   | null                                   | int    | 0                                         |
| struct  | Struct<br>Alle Members sind default(T) | long   | 0L                                        |
| bool    | false                                  | sbyte  | 0                                         |
| byte    | 0                                      | short  | 0                                         |
| char    | '\0'                                   | uint   | 0                                         |
| decimal | 0.0M                                   | ulong  | 0                                         |
| double  | 0.0D                                   | ushort | 0                                         |
| float   | 0.0F                                   | enum   | Resultat aus (E)0<br>E = Enumerations-Typ |

Kann auch im Code verwendet werden

Sonst NULL

```
int x = default(int);
```

```
int x = default;
```